Università degli Studi di Milano

Corso di Laurea in
Sicurezza dei Sistemi e delle Reti Informatiche

Lezione 3 – Algoritmi di matching e tassi di errore

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Sitemi biometrici

Modulo 3 - Unità Didattica 2

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1. Introduction

In this section we introduce the problem of matching two fingerprints, we resume the most important critical factors and the available countermeasures that are mostly used in the applications.

1.1 Problem definition

Given two different *fingerprints* (hereafter: FPs), matching them means to give an answer to the following question: *do they come from the same person*?

In FP matching, ***four*** different cases can take place:

- FPs actually DO come from the same person and the AFIS declares they DO (we refer to this case as ***Correct Match***)
- FPs actually DO come from the same person and the AFIS declares they DO NOT (***False Non-Match***)
- FPs actually DO NOT come from the same person and the AFIS declares they DO (***False Match***)
- FPs actually DO NOT come from the same person and the AFIS declares they DO NOT (***Correct Non-Match***)

In brief, we have two case of errors (***False Match*** , ***False Non-Match***) and two correct answers (***Correct Match*** and ***Correct Non-Match***). Depending on the application, the errors can have a different risk and different costs.

1.2 A complex task to achieve

Fingerprint matching is an ***extremely difficult problem***, mainly due to the large variability in different impressions of the same finger (large intra-class variations). As stated in [1]: "here comes a stage in the development of any biometric recognition system where it becomes increasingly difficult to achieve significantly better performance from a given biometric identifier […]. There is a popular misconception that automatic fingerprint matching is a fully solved problem since it was one of the first applications of automatic pattern recognition. Despite notions to the contrary, there are a number of challenges that remain to be overcome in designing a completely automatic and reliable fingerprint matcher, especially when fingerprint images are of poor quality".

1.2.1 Negative factors in fingerprint matching

There are several negative factors (i.e.: dirt on sensing device, poor user attitude, cuts, scars or burns on fingertip surface, sweat, either too weak or too strong pressure on sensing device, human skin elasticity) that can influence the acquisition phase producing different kinds of errors:

- spurious features may be introduced;
- genuine features may be discarded or lost.

This consequently implies some problems, as:

- we may not have an exact match (YES/NO type);
- the input image (the one coming from the person who is trying to access the system) can be substantially different from the template image (the one stored in the DB) even though they represent the same finger.

Moreover, as seen in previous lessons, different problems may arise (Figure 1):

- the input image may be noisy and distorted (dust on the sensor, not correct finger pressure on the sensor);

- different skin conditions can occur (dryness, cuts, scratches , etc.);
- The input image may contain only a part of the whole FP (due to the small acquisition area of certain sensing devices)

Figure 1. Matching algorithms must face hard operative conditions such as: (a) different skin conditions during acquisition; (b) small area of overlap.

1.2.2 Countermeasures

As countermeasures, it is possible to model the differences between input and template images and try to use the model to reduce the effects of the negative factors on the matching algorithms. Differences that can be modeled are:

- linear transformation (Rotation and/or Translation);
- non-linear deformation (the act of detect deformation maps of the shape of the finger onto the two-dimensional surface of the sensor).

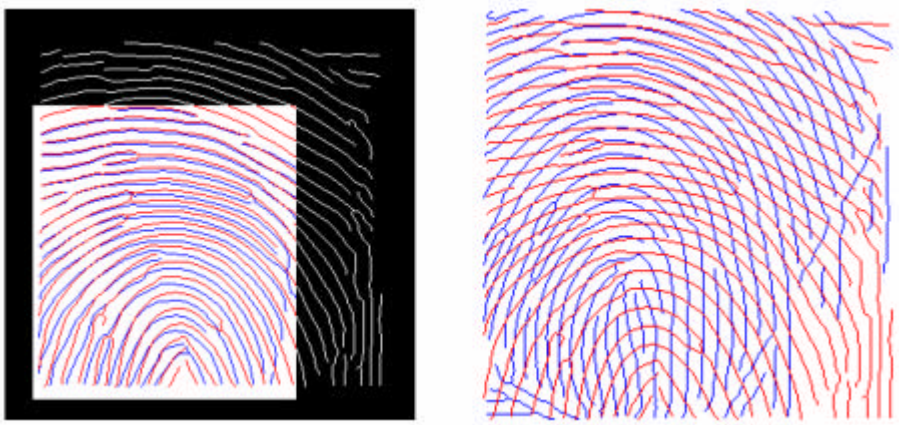


Figure 1.1 Reduction of the rotation and translation differences between two ridge patterns before to achieve the real matching. (a) Alignment of two impressions of the same finger (the non-overlapping regions appear as white on black); (b) Alignment of two impressions of different fingers.

Figure 1.1 shows the aligning of the input image with the template image. The red thinned image is the template, and the blue thinned image is the input. This phase is commonly used to reduce the rotation and translation problems before to achieve the proper matching.

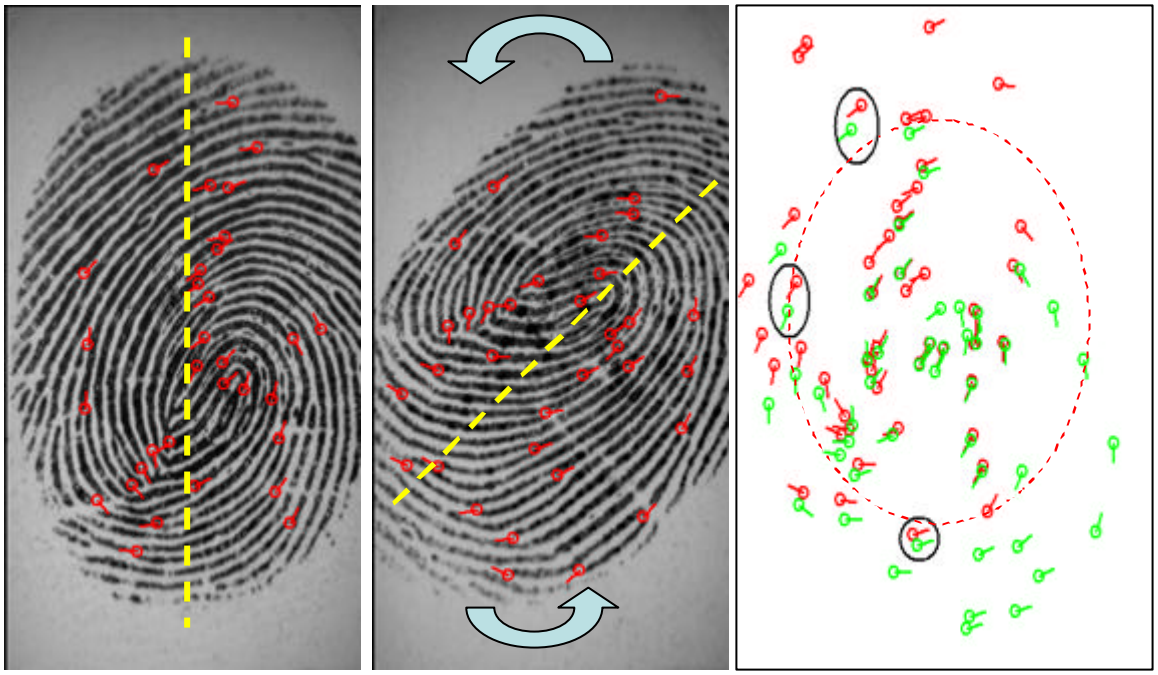


Figure 1.2. Rotation and translation required to perform a correct match between two fingerprints. Far from the center, the deformation distortion is more valuable: the distance between couples of minutiae tend to increase.

In the literature, many algorithms capable to reduce the effect of the skin deformation during acquisition are present (Figure 1.2). These algorithms tend to reduce the intra-class distance between FPs coming from the same individual (deformation), but they typically decrease also the inter-class distance between FPs coming from different users.

2. How Matching works

Matching two fingerprints means to compare two fingerprints images (one *input* fingerprint and one *reference* fingerprint) -or their features- and to calculate their "similarity level". The measure for the similarity is the so-called "matching score", which generally assumes values between 0 (no similarity) and 100 (complete identity).

Sometimes, instead of the similarity, it is calculated some kind of "distance measure" between input and reference fingerprint. The greater the similarity, the smaller the distance between FP features.

In our formulation, a given pair of feature sets is considered to be *extracted from the same individual* if the computed *matching score is greater than a fixed threshold*.

In general, fingerprint matching paradigms can be partitioned in the following categories [2]:

- **Manual**.
  - Human experts use a combination of visual, textural, minutiae cues and *experience* for verification.
  - This system is still used in the final stages of law enforcement applications
- **Image based** (Correlation based).
  - It utilizes only visual appearance.

  - It requires the complete image to be stored (large template sizes)

- **Texture/Ridge based** (Correlation based).
  - It treats the fingerprint as an oriented texture/ridge image.
  - It is less accurate than minutiae based matchers since most regions in the fingerprints carry low textural content

- **Minutiae based**
  - It uses the relative position of the minutiae points.
  - It's the most popular and accurate approach for verification.
  - It resembles manual approach very closely.

The *minutiae based approach* is probably the most used in the literature and in the commercial applications. However, there are some difficulties when using the minutiae approach. It is difficult to accurately extract the minutiae points when the fingerprint image is of low quality. Furthermore, this method does not take into account the global pattern of ridges and furrows.

The correlation-based method is able to overcome some of the difficulties of the minutiae-based approach. However, it has also some shortcomings. Correlation-based techniques require the precise location of a registration point and are affected by image translation and rotation.

Fingerprint matching based on minutiae has problems in matching different sized (unregistered) minutiae patterns. Local ridge structures can not be completely characterized by minutiae.

A commercial fingerprint-based authentication system requires a very low False Reject Rate (FAR) for a given False Accept Rate (FAR). This is very difficult to achieve with any one technique. The usage of different matching techniques can increase the overall accuracy of the system. In a real application, the sensor, the acquisition system and the variation in performance of the system over time is very critical. We will review the available solutions to this problem in the module of the course dedicated to the multimodal biometrics.

2.1 Image Based Matching: Optical Correlation

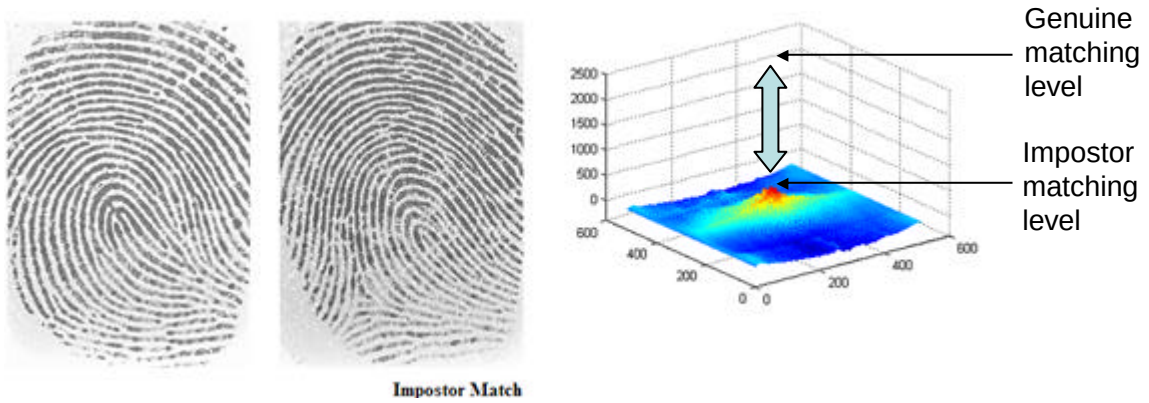


Figure 2.1. Direct Optical correlation of fingerprint images.

This technique simply performs a correlation between the two images. The correlation function present a maxim values that depends on if the two fingerprint comes from a genuine match or from a impostor match

Let us resume the advantages and disadvantages of this technique.

Advantages:

- Image itself is used as the template
- Requires only low resolution images
- Optical correlation makes it extremely fast (Choudary and Awwal '99, Lee et al. 99, Roberge et al. 99, Baze et al.00)

Disadvantages:

- Image itself is used as the template (template size about 30 KB)
- Requires accurate alignment of the two prints (unreliable in poor prints)
- Not robust to changes in scale, orientation and position.

2.2 Texture Based Matching: Filterbanks

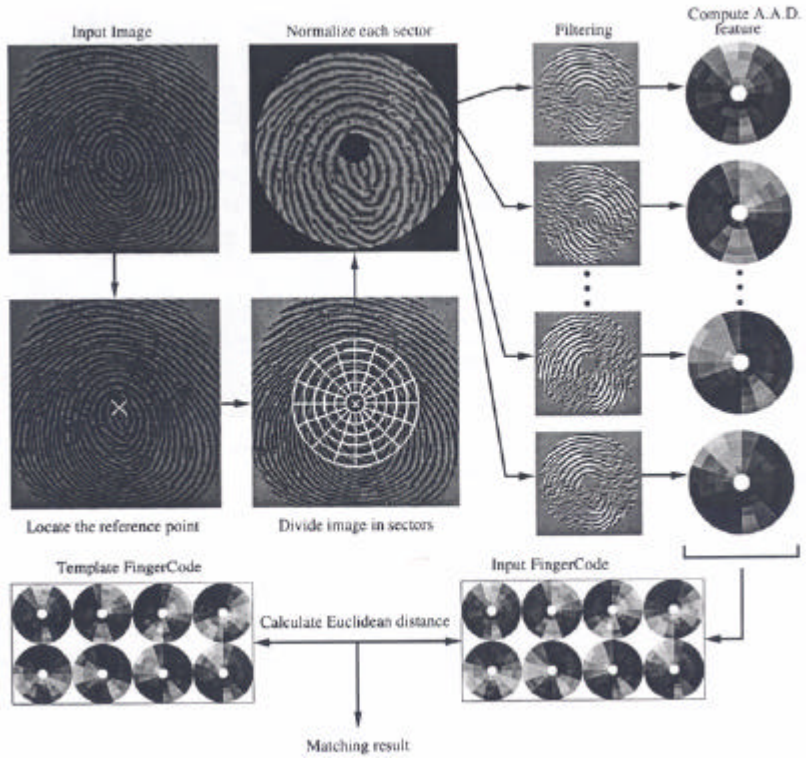


Figure 2.2. Texture Based Matching: Filterbanks

This technique extracts local information in N small region of the image. Using a Gabor filter bank of M filters, it extracts M coefficients/filtered images for the each of the N regions processed. The matching algorithm compares the MxN items between the two images.

Let us resume the advantages and disadvantages of this technique.

Advantages:

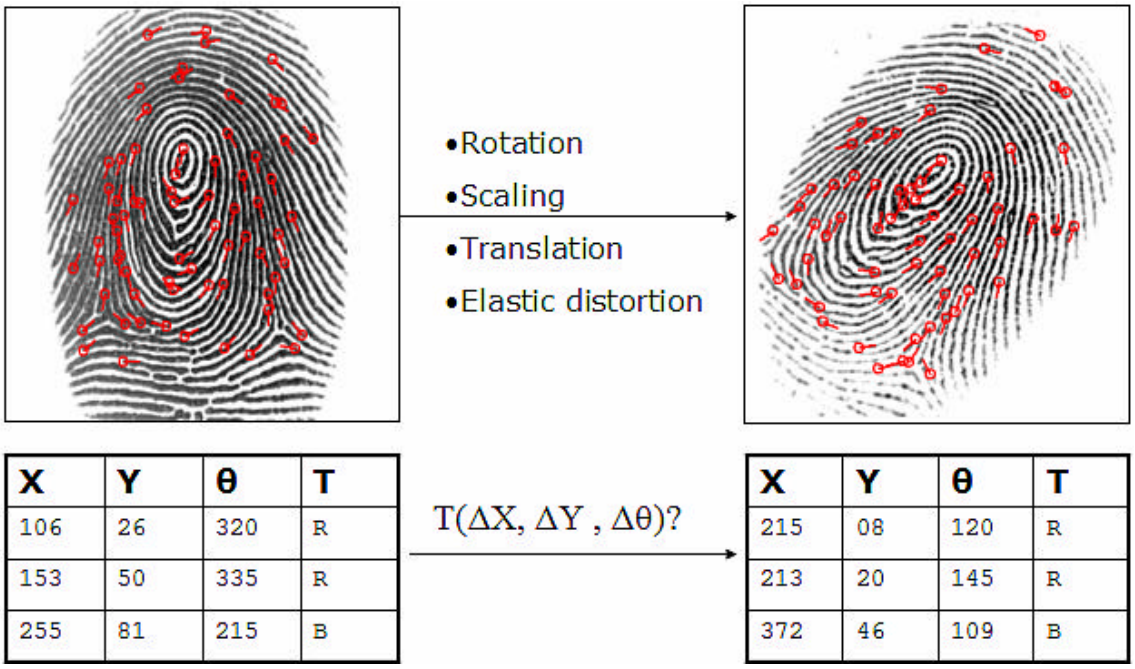
- Uses texture information (lost in optical and minutiae based schemes)
- Performs well with poor quality prints

- Features are statistically independent from minutiae and can be combined with minutiae matchers for higher accuracy

Disadvantages :

- Requires accurate alignment of the two prints (unreliable in poor prints)
- Not invariant to translation, orientation and non-linear distortion.
- Less Accurate than minutiae based matchers

2.3 Minutiae Based Matching



X	Y	θ	T
106	26	320	R
153	50	335	R
255	81	215	B

X	Y	θ	T
215	08	120	R
213	20	145	R
372	46	109	B

Figure 2.3. Minutiae based matching. X and Y are the pixel coordinates of the extracted minutiae; ? represents the orientation angle of the minutia, T the type of the minutia (**B**ifurcation and end-**R**idge).

Minutiae-based techniques firstly find the minutiae points in the two finger image to compare (the coordinates tables in Figure 3.3). In order to achieve a correct minutiae comparison, some preprocessing is often required:

- Rotation alignment;
- Scaling alignment
- Translation alignment
- Elastic distortion reduction.

Once the input image is properly rotated, translated and scaled, the matching algorithm can start to compare the minutiae.

A Fingerprint Matching problem is substantially a point-pattern matching problem. Matching of two fingerprints means comparing two sets of points (minutiae), one belonging to the template FP and the other belonging to the input FP. A method to represent minutiae of a FP image is storing them in a matrix. Here is the mathematical formalism:

$M = (m_1, m_2, \Lambda\ , m_N)^T$, where N in the number of minutiae;

$m_i = (x_i, y_i, q_i, t_i)$, where x and y are the coordinates of the i-th minutia, q is its orientation angle and t the minutia type (either ridge bifurcation or ridge end).

Let us resume the advantages and disadvantages of this technique.

Advantages:

- Invariant to translation, rotation and scale changes
- Very accurate

Disadvantages:

- Minutiae extraction is error prone is low quality images
- Not robust to non-linear distortion.
- Does not use visual and textural cues

Minutiae-based matching can be approached in different ways (for instance: *core-based* approach, *global structure* approach and the *local structure* approach).

2.3.1 Core-based approach

Core Point is used to align the two FP images. Most of the non-core-based matching algorithms are time consuming. Therefore, they are not suitable for on-line applications. On the contrary, core-based matching algorithms are more efficient than non-core-based algorithms, but they highly depend on core detection precision. This is a difficult task on poor quality images.

2.3.2 Local structure approach

Advantages. These techniques are *Rotation* and *translation invariant* because they consider only of the *difference* in the directions and the locations of the minutiae. Moreover, local structure approach can tolerate reasonable deformation since these methods process only small areas of the fingerprint at time.

Disadvantages. Prints from different fingers may have quite a few similar local structures and prints from the same finger (small overlap area). In addition it can happen to have only few similar structures to compare due to the presence of spurious minutiae and the absence of genuine ones. Therefore, FP matching based only on local structure is less reliable.

2.3.3 Global structure approach

Advantages. Reliable determination of the uniqueness of fingerprints.

Disadvantages. The non-linear deformation of a FP image increases as the distance from the reference minutia for alignment increases (Figure 1.2). Therefore, the performance of matching algorithms using only global alignment is less reliable. The global features of the minutiae are dependent on the rotation and translation of the fingerprint. It cannot be directly used for matching, image preprocessing is typically required.

2.3.4 Hybrid structure approach

This approach makes use of both local and global features taken from FP images. First, local features are used to locate the “best matched minutia pair”, one in the input FP and the other in the template FP. Secondly, the best

matched local structures provide the correspondences for aligning the two FPs. Finally, the "similarity level" is computed and compared to the threshold in order to decide the matching result.

Figure 2.4 shows the case where a group of 5 minutiae is selected for comparison in the reference fingerprint.

Other hybrid approaches control if two region of a fingerprint are similar by controlling if the triangles between the minutiae are similar checking the angles, the distances measured using ridge numbers and the types of the minutiae.

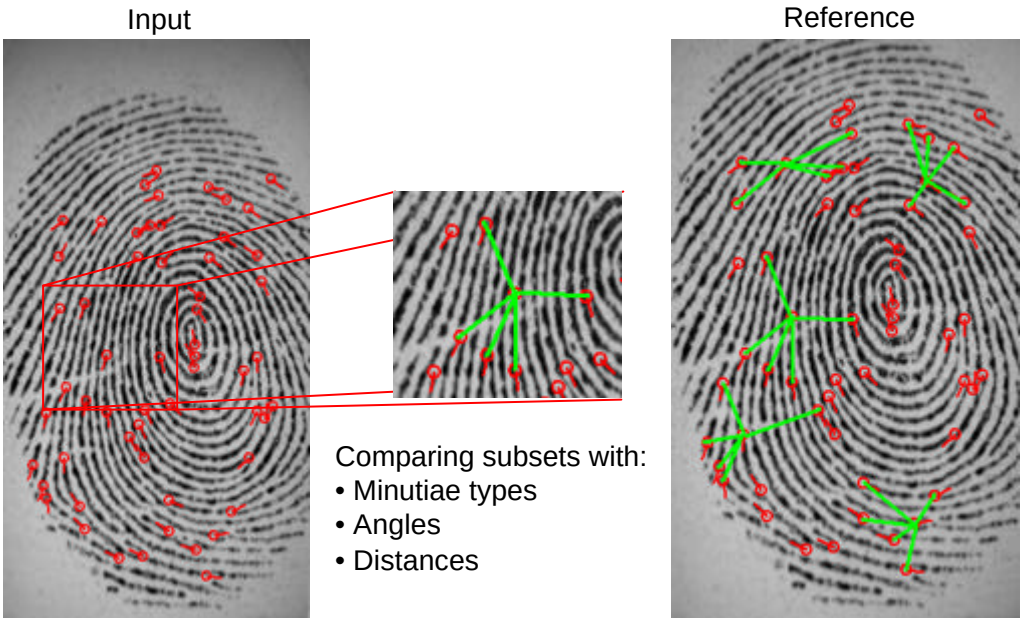


Figure 2.4. Hybrid matching. Local (minutiae) and global features (ridge counts, angles between minutiae) are used to compare the presence of subsets of K=5 minutiae present in the input image in the reference image.

Figure 2.5. The Kovac matching algorithm.

3. State of the art

TABLE I – Average Fingerprint Accuracy on Standard Databases				
FTE Failure to enroll	FNMR False Non Match Rate	FMR1 Verification error rate	FMR2 Projected False Match rate over 1 million people	FMR3 Projected False Match rate over 500 people
4%	2%	2%	0.001%	<1%

In this section we describe the available results present in the state of the art in fingerprint matching systems. The performances index has been processed using standard datasets of fingerprints. Two main public archives of fingerprints are available. A first group of dataset is produced by the US National Institute of Standard Technology (NIST), and a second group is produced by the international competition on fingerprint verification organization FVC2004. The NIST DBs are mainly focused on off-line fingerprint. FVC2004 uses only life scan fingerprint images. Table I presents the results published in "Biometrics: A Grand Challenge" by Anil K. Jain et. Al.

3.1 FVC2004

The FVC2004 competition presented four different databases (DB1, DB2, DB3 and DB4) collected by using the following sensors/technologies:

- DB1: optical sensor "V300" by CrossMatch
- DB2: optical sensor "U.are.U 4000" by Digital Persona
- DB3: thermal sweeping sensor "FingerChip FCD4B14CB" by Atmel
- DB4: synthetic fingerprint generation

Execution times and memory occupations has been evaluated on a standard hardware and software architecture. Here it follows the characteristics of the sensors used to build the DBs.

	Sensor Type	Image Size	Set A (wxd)	Set B (wxd)	Resolution
DB1	Optical Sensor	640x480 (307 Kpixels)	100x8	10x8	500 dpi
DB2	Optical Sensor	328x364 (119 Kpixels)	100x8	10x8	500 dpi
DB3	Thermal sweeping Sensor	300x480 (144 Kpixels)	100x8	10x8	512 dpi
DB4	SFinGe v3.0	288x384 (108 Kpixels)	100x8	10x8	about 500 dpi

In this competition, the Set B (8 fingers for 10 people) has been released *before the competition* to tune the candidate systems. Please note that the real test set (Set A) has been released *after the submission of algorithm* to the organization. That can be considered a "validation" test, using the Pattern Matching vocabulary.

Here it follows the average results over all databases.

Algorithm	EER(%)	AvgEnroll Time (sec)	AvgMatch Time (sec)	AvgModel Size (KB)
BioscryptInc.	2.07	0.08	1.48	24
SondaLtd	2.10	2.07	2.07	1.3
Chinese Academy of Sciences	2.30	0.35	0.67	16.4

Gevarius	2.45	0.69	0.71	2.0
Jan Lunter	2.90	1.01	1.19	3.1

3.2 The Fingerprint Vendor Technology Evaluation (FpVTE) 2003

The Fingerprint Vendor Technology Evaluation (FpVTE) 2003 is an independently administered technology evaluation of fingerprint matching, identification, and verification systems. FpVTE 2003 is being conducted by the National Institute of Standards & Technology (NIST) on behalf of the Justice Management Division (JMD) of the U.S. Department of Justice.

FpVTE was designed to assess the capability of fingerprint systems to meet requirements for both large-scale and small-scale real world applications. FpVTE 2003 consists of multiple tests performed with combinations of fingers (e.g., single fingers, two index fingers, four to ten fingers) and different types and qualities of operational fingerprints (e.g., flat livescan images from visa applicants, multi-finger slap livescan images from present-day booking or background check systems, or rolled and flat inked fingerprints from legacy criminal databases).

FpVTE 2003 will serve as part of NIST's statutory mandate under section 403c of the USA PATRIOT Act to certify those biometric technologies that may be used in U.S. VISIT (U.S. Visitor and Immigrant Status Indication Technology, formerly known as the US entry-exit system). In addition, FpVTE results may form the basis for the design and acquisition of other large-scale fingerprint identification systems. pVTE 2003 evaluations were conducted at the NIST facilities at Gaithersburg, MD, from October 2003 through November 2003.

Small, Medium, and Large-Scale Tests (SST, MST and LST) were conducted using systems provided by 18 Participants: 123 ID, Inc. (MST, LST) , Antheus Technology, Inc. (MST, LST) , Av@lon Systems Inc. (MST) , BioLink Technologies International, Inc. (MST, LST) , Bioscrypt, Inc. (SST) , Cogent Systems, Inc. (SST, MST, LST) , Dermalog Identification Systems GmbH (MST, LST) , Golden Finger Systems (MST, LST) , Griaule Tecnologia (LST) , Identix, Inc. (MST, LST) , SAGEM MORPHO, Inc. (MST, LST) , NEC (MST, LST) , Neurotechnologija Ltd. (MST) , The Phoenix Group, Inc. (MST) , Motorola (MST, LST) , Raytheon Company (MST, LST) , Technoimagia Co., Ltd. (MST) , Ultra-Scan Corporation (MST),

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The most accurate fingerprint system tested (NEC MST) using operational quality ***single*** fingerprints:

- 99.4% true accept rate @ 0.01% false accept rate
- 99.9% true accept rate @ 1.0% false accept rate

The complete tests are described in the appendix.

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4. A complete example of matching technique

In the following paragraphs we give a brief description of one hybrid approach method. The aim of this section is to give an idea of the complexity of the algorithms, not to be an exhaustive description; all the details can be found in the cited papers published by IEEE.

4.1 Fingerprint Minutiae Matching Based on the Local and Global Structures

This method has been proposed by Xudong Jiang and Wei-Yun Yau in the 15th Pattern Recognition International Conference, September 2000.

Algorithm Description

1. A control on the number of extracted minutiae is performed. The "image center" is identified; the image center is assumed to be the mass-center of the detected minutiae. If the number of minutiae is greater than a certain threshold established in advance, only N minutiae are selected, according to their distance from mass-center.

2. The distances between each minutia in the FP image and all the other minutiae are computed; the results are stored in the "distance matrix".

3. It is possible to describe a minutia with rotation and translation invariant features by using some other minutiae in its neighbourhood. For each minutia:

4. The relative distance,

5. The radial angle,

6. The minutia direction between minutia and its k-nearest neighbours are computed.

7. So, the Local Features Vector (LFV) for each minutia is computed. It is an 11-element array. The 2-nearest neighbours have been considered in this algorithm.

8. The difference vectors between each input FP LFV and each template FP LFV are calculated.

9. A Similarity Level for each minutia pair is computed, only if the difference vector is lower than a certain threshold; the similarity level $sl(i, j)$ is higher than 0 and lower than 1; it describes a matching certainty level of a local structure pair instead of simply matched or not matched. $sl(i, j) = 1$ implies a perfect match, while $sl(i, j) = 0$ implies a total mismatch.

10. If all the elements in the Similarity matrix are equal to zero, it is possible to declare that input and template FPs do NOT match.

11. Otherwise, a minutiae pair is considered as reference to align the two images; the "best-matched pair", one from the input FP and another from the template FP, provides a reliable correspondence of the two FPs. All other minutiae will be aligned based on this corresponding reference minutiae pair by converting them to polar coordinate system with respect to the best-matched pair.

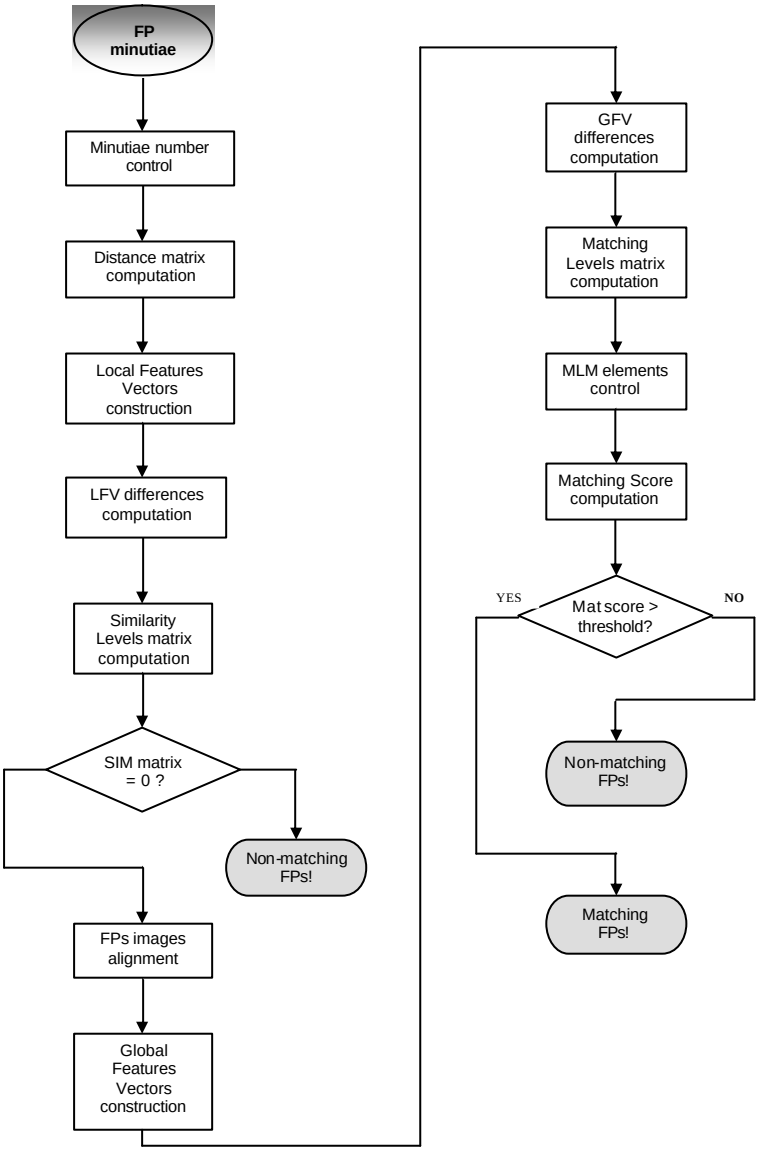
12. For each minutia in the image, except for the reference one, a Global Features Vector (GFV) is computed; this is a 3-element vector containing:

13. The distance to the alignment minutia,

14. The radial angle,

15. The angle difference between considered minutia and reference minutia.
16. The difference vectors between each input FP GFV and each template FP GFV are calculated.
17. A Matching Level for each minutia pair is computed, only if the difference vector is lower than a certain threshold; the matching level $ml(i, j)$ is higher than 0 and lower than 1; $ml(i, j) = 1$ implies a perfect match, while $ml(i, j) = 0$ implies a total mismatch.
18. To avoid a minutia being doubly used for matching, $ml(i, j)$ is set to zero if there exists any k, such that ml(i, k) > ml(i, j) or ml(k, j) > ml(i, j).
19. A Matching Score is computed; it represents the average matching certainty level of the minutiae extracted in the FP image.
20. The matching score is compared to a certain threshold established in advance; if the matching score is greater than threshold, then the input and the template FPs match.

This algorithm presents low computational complexity, but on the contrary the proposed algorithm suffers when a large number of false (spurious) minutiae are extracted. In the following page a block-diagram of the described algorithm is shown.



5. References

[1] D. Maltoni, D. Maio, A. K. Jain, S. Prabhakar, "Handbook of Fingerprint Recognition", Springer, 2003
[2] Sharat Chikkerur, "Online Fingerprint Verification", Report of the Center for Unified Biometrics and Sensors, University at Buffalo
[3] V.S. Valencia, 'Biometric Testing: It's Not as Easy as You Think', Biometric Consortium Conference, Sept. 2003, Arlington VA USA.
[4] A. J. Mansfield, J. L. Wayman, "Best practices in testing and reporting performance of biometric devices", Version 2.01, Aug. 2002.

6. Appendix (Results of the FPVTE2003 evaluation)

FPVTE2003

Test	Compares	# Subtests	# Comparisons	# Systems Successfully Completed	Allowed Time
LST	Sets of 1-10 fingerprint images (Flat, Slap, and Rolled; various combinations of fingers)	31 (uses 10 datasets containing 64,000 fingerprint sets)	1.044 billion set-to-set comparisons	13	21 days
MST	Single images (Flat & Slap Right index)	1 (compares a single 10,000 image dataset against itself)	100 million single image comparisons	18	14 days
SST[2]	Single images (Flat Right index) (Subset of MST)	1 (compares a single 1,000 image dataset against itself)	1 million single image comparisons	3 (SST only) 21 (as a subset of MST)	14 days

The following results has been processed with FAR = 0.0001 and plotting the True Accept Rate

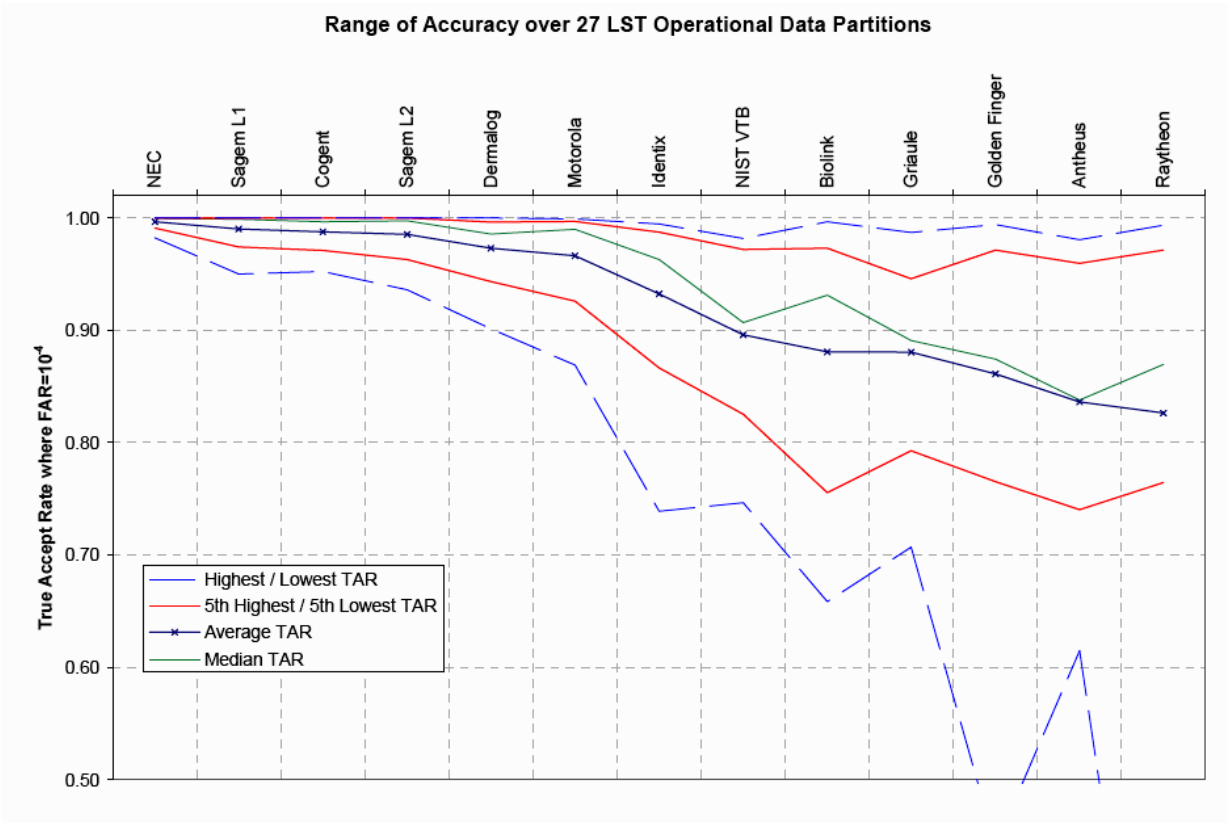


Figure 1. Range of Accuracy over 27 Operational LST Partitions. The systems are sorted by their average accuracy over the 27 partitions; note that sorting by median performance would change the order for some systems.

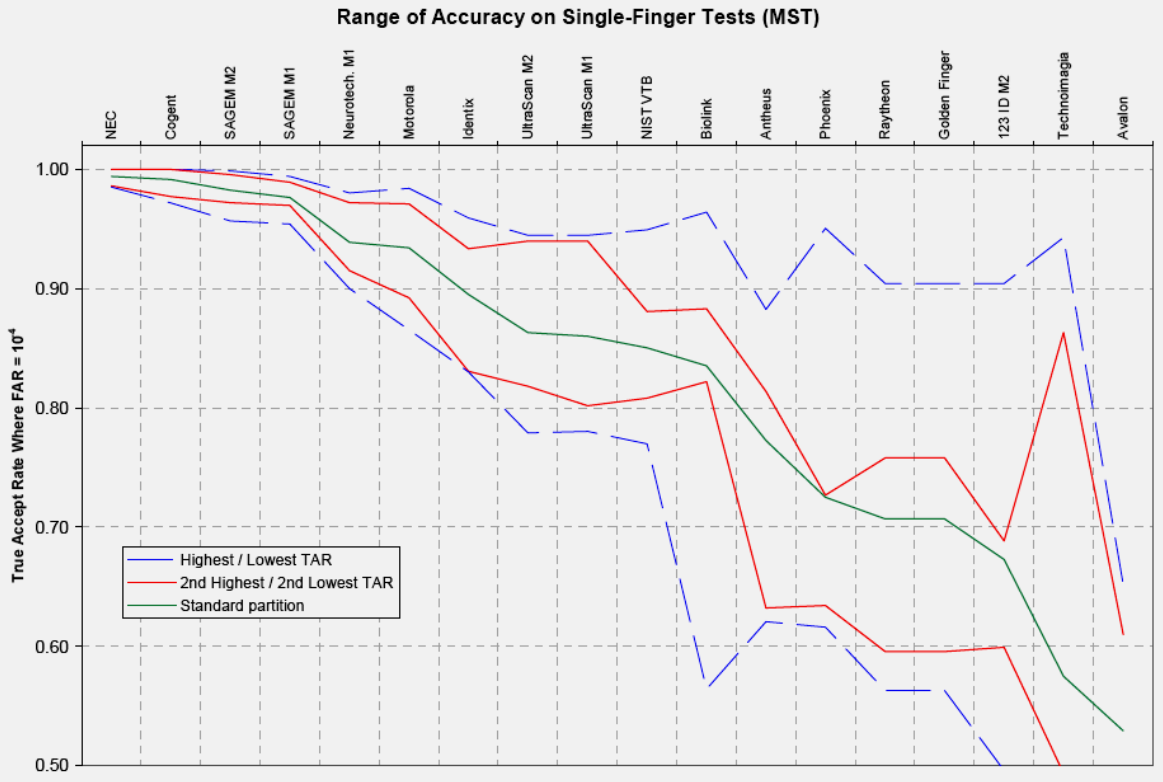


Figure 2. Range of accuracy across 7 MST partitions. These systems are sorted by the systems' performance on the standard MST, which is simply the combination of the seven partitions.[1] The large difference between the highest and second highest TARs is attributable to the presence of data collected in a controlled (highest) versus operational (second highest) setting.

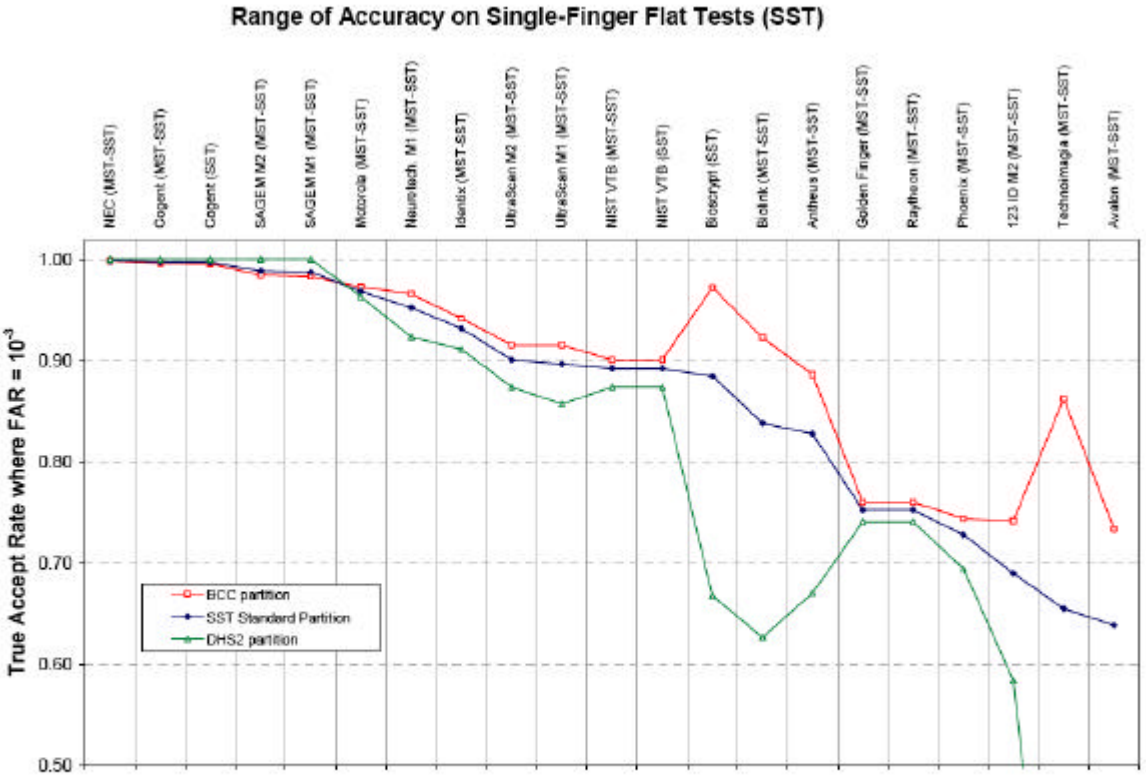


Figure 3. Range of Accuracy on Single-Finger Flats (SST). These systems are sorted by performance on the SST standard partition. Note that these results are reported at FAR=0.1%, in contrast to most of the figures in this report, which are based on FAR=0.01%.

All documents are available at http://fpvte.nist.gov/